

UX4G Design System Migration

Technical Migration Blueprint

SARGAM 2.0

853

Blade templates
audited

537

Page views in scope

1,572 h

Estimated effort (P50)

~5.2 mo

With 3 FE + 1 QA

PREPARED FOR	Sargam 2.0 development team & management
SCOPE	Frontend architecture only — no backend, route, API or schema change
TARGET	UX4G Design System v2.0.8 (NeGD / MeitY)
AUDIT BASIS	Working tree at HEAD <code>60715da05</code> + live UX4G CDN artefacts
DATE	2026-08-02

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Prepared for: LBSNAA / Sargam 2.0 development team **Scope:** Frontend architecture and reusable components only. No backend, route, controller, API, schema or business-logic change. **Audit basis:** Static analysis of the working tree at branch `.invalid` (HEAD `60715da05`), plus live inspection of the UX4G CDN artefacts.

Evidence rule applied throughout. Every number in this document was measured from the codebase or downloaded from `cdn.ux4g.gov.in`. Where a fact could not be confirmed it is explicitly tagged `Needs Code Inspection` rather than estimated.

1. Executive Summary

1.1 The headline finding

UX4G v2.0.8 is a rebranded recompilation of Bootstrap 5.3 — not a different framework. This was verified by downloading and inspecting the actual CDN artefacts:

Probe	Result	Implication
<code>cdn.ux4g.gov.in/UX4G@2.0.8/js/ux4g.min.js</code> header	UX4G v2.0.8 ... Licensed under MIT	Distinct product
UMD export in that file	<code>(t=globalThis...).bootstrap = e(t.Popper)</code>	Registers the global as <code>bootstrap</code> , not <code>UX4G</code>
Event namespaces in that file	<code>.bs.modal .bs.dropdown .bs.collapse .bs.tab .bs.offcanvas .bs.toast .bs.tooltip .bs.popover .bs.carousel .bs.scrollspy .bs.alert .bs.button .bs.backdrop</code>	Identical to Bootstrap 5
<code>ux4g-min.css</code> custom properties	<code>--bs-primary</code> , <code>--bs-body-bg</code> , <code>--bs-border-radius</code> , ...	Bootstrap's <code>--bs-*</code> token namespace , not <code>--ux4g-*</code>
Class surface in <code>ux4g-min.css</code>	<code>.btn-primary .card-body .form-control .form-select .form-check-input .modal-dialog .offcanvas .accordion-button .nav-tabs .dropdown-menu .breadcrumb .pagination .badge .toast .tooltip .popover .carousel .progress .spinner-border .list-group .input-group .navbar</code> all present	Full Bootstrap component parity
Grid classes present	60 distinct <code>.col-{bp}-{n}</code>	Bootstrap grid, unchanged
Spacing utilities present	84 distinct <code>.{m,p}{t,b,s,e,x,y}-{0..5}</code>	Bootstrap spacing scale, unchanged
Breakpoints	576 / 768 / 992 / 1200 / 1400 px	Identical to Bootstrap 5
Documented modal markup	<code>data-bs-toggle="modal"</code> , <code>data-bs-dismiss="modal"</code> , <code>.modal-dialog-centered</code>	Identical attribute contract

Consequence: the migration is **not** a component rewrite. The 2,202 `btn btn-*` , 2,288 `card` , 1,697 `form-control` and 350 `data-bs-toggle` usages in this codebase **do not need to be renamed**. This inverts the normal cost model of a design-system migration.

1.2 What the project actually costs

The expensive work is *not* the components. It is four things:

- Reconciling ~1.7 MB of accumulated custom CSS** (7,128 `!important` declarations) that currently overrides the existing Bootstrap theme and will fight the UX4G theme identically.
- A brand-colour decision:** the portal's `--bs-primary` is `#004384` (LBSNAA navy); UX4G ships `#613AF5` (violet). This is a governance decision, not a technical one, and it gates the entire visual sprint plan.
- Consolidating four coexisting icon systems** into one.
- Visual QA across 494 admin pages** — the dominant line item.

1.3 Recommended strategy

Token-overlay adoption, not a rip-and-replace. Because the class contract is identical, UX4G can be introduced as the *base* stylesheet under the existing theme, with the project's overrides progressively deleted rather than rewritten. This preserves a working application at every commit and makes rollback a one-line layout change.

1.4 Headline numbers

Metric	Value
Blade files	853
Page views (files with <code>@extends</code>)	537
Pages on <code>admin.layouts.master</code>	494 (92 %)
Custom + theme CSS shipped	~1.70 MB (<code>public/css</code> 505 KB + <code>admin_assets/css</code> 1,198 KB)

Metric	Value
<code>!important</code> declarations	7,128
Inline <code><style></code> blocks	379 across 319 files
Inline <code>style=""</code> attributes	2,956
Vendored JS libraries	35
jQuery <code>\$(...)</code> call sites in Blade	5,851 across 192 files
Distinct external CDN hosts	25
Estimated effort (P50)	1,572 h (range 1,180 – 2,100)
Recommended team	3 frontend + 1 QA + 0.4 designer, ~5 months

1.5 Three risks that must be decided before Sprint 1

#	Decision	Why it blocks
R-1	Brand colour: adopt UX4G violet <code>#613AF5</code> , or retain LBSNAA navy <code>#004384</code> as an approved token override?	Determines whether Sprint 2–4 is a re-theme or a no-op. Affects every screenshot in UAT.
R-2	Version pinning: only <code>UX4G@2.0.8</code> resolves on the CDN (<code>2.1.0</code> , <code>3.0.0</code> , <code>latest</code> all return <code>404</code>). The docs advertise "v3.0".	We must either pin 2.0.8 and self-host, or obtain 3.x from NeGD.
R-3	Self-hosting: the CDN's <code>ux4g-min.css</code> declares <code>@font-face</code> at <code>../fonts/NotoSans-*.woff2</code> — every one of those URLs returns 404 , as do the five <code>../img/*-icons.svg</code> sprites.	Consuming UX4G purely by CDN yields no Noto Sans and no gov icon sprites . Self-hosting is mandatory, not optional.

2. Current Architecture

2.1 Stack (confirmed)

Layer	Technology	Version	Source
Framework	Laravel	^9.0	<code>composer.json</code>
PHP		^8.0	<code>composer.json</code>
Templating	Blade	—	853 files
CSS framework	Bootstrap	5.3.2 (bundled) / 5.3.3 (<code>node_modules</code>) / 5.3.6 (CDN, timetable layout)	see §2.4
Build	Laravel Mix (webpack)	^6.0.6	<code>package.json</code> ; a <code>vite.config.js</code> also exists but Mix is what <code>npm run prod</code> invokes
CSS preprocessor	Sass	^1.56.1	<code>resources/sass/</code>
DOM library	jQuery	3.7.1 (CDN)	<code>public/js/jquery-3.7.1.min.js</code> is 0 bytes — jQuery is served from <code>code.jquery.com</code>
Admin theme	Modernize-family Bootstrap 5 admin template	—	<code>public/admin_assets/</code>
Server-side tables	Yajra DataTables	^10.0	55 PHP DataTable classes in <code>app/</code>
Permissions	spatie/laravel-permission	^6.16	drives the dynamic RBAC sidebar

2.2 Layout inheritance

Layout	Pages	Lines	Notes
<code>admin.layouts.master</code>	494	1,318	The primary target. Renders a 5-tab pane system with server-resolved RBAC sidebar.
<code>fc.layouts.master</code>	31	25	Clean, modern, already has a <code>skip-link</code> target and <code><main id="content" tabindex="-1"></code> . Best-practice reference.
<code>layouts.app</code>	7	—	Laravel/UI auth scaffolding.
<code>admin.layouts.timetable</code>	4	—	Standalone — loads its own Bootstrap 5.3.6 from jsDelivr.
<code>faculty.layouts.master</code>	1	674	Near-dead; consider retiring.

Architectural note (from `master.blade.php:62–91`): the master layout resolves which of five `@section` panes a page actually populated, and renders it into the active RBAC tab. This coupling between the sidebar resolver and the content pane is **fragile and must not be refactored during the UX4G migration** — it is orthogonal to styling. Treat it as frozen.

2.3 Stylesheet load order (`admin/layouts/pre_header.blade.php`)

1. <code>admin_assets/css/styles.css</code>	← 1,002 KB, 5,579 !important	(theme + Bootstrap build)
2. <code>css/custom.css</code>	← 69.7 KB, 209 !important	
3. <code>css/admin-header.css</code>	← 22.9 KB, 195 !important	
4. <code>admin_assets/css/dashboard-enhanced.css</code>		
5. <code>datatables.net-bs5.css</code>		(local)
6. <code>datatables responsive 2.5.0</code>		(CDN)
7. <code>datatables buttons 2.4.2</code>		(CDN)
8. <code>admin_assets/css/material-icons-local.css</code>		
9. <code>css/spacing-system.css</code>	← 4.8 KB, 42 !important	
10. <code>bootstrap-icons 1.11.3</code>		(CDN)
11. <code>css/breadcrumb.css</code>		
12. <code>css/sidebar-menu-enhanced.css</code>		
13. <code>admin_assets/css/sidebar-modern.css</code>	← 22.2 KB, 277 !important	
14. <code>css/sargam-app.css</code>	← 50.9 KB, 32 !important	← the sanctioned design-system file

This ordering is the crux of the migration. UX4G must be inserted at position 0 (or replace position 1), and layers 2–13 must be progressively deleted. `sargam-app.css` is the correct long-term home for everything that survives.

2.4 Version drift (defect, pre-existing)

Three Bootstrap versions are live in one application:

Where	Version
public/admin_assets/libs/bootstrap/dist/js/bootstrap.bundle.min.js	5.3.2
node_modules/bootstrap	5.3.3
admin/layouts/timetable.blade.php:305 (jsDelivr)	5.3.6

Resolve this **before** Sprint 1; otherwise migration bugs will be indistinguishable from version-skew bugs.

3. Bootstrap Inventory

3.1 Component classes in Blade

Counts are `class="..."` occurrences across `resources/views/**/*.blade.php`.

Bootstrap component	Occurrences	Rank
<code>card</code>	2,288	1
<code>btn btn-*</code>	2,202	2
<code>form-control</code>	1,697	3
<code>table</code>	997	4
<code>form-select</code>	861	5
<code>form-check</code>	811	6
<code>card-body</code>	789	7
<code>badge</code>	600	8
<code>alert alert-*</code>	384	9
<code>modal-dialog</code>	207	10
<code>modal fade</code>	207	11
<code>breadcrumb</code>	135	12
<code>spinner-*</code>	95	13
<code>collapse</code>	94	14
<code>input-group</code>	92	15
<code>dropdown-menu</code>	59	16
<code>navbar</code>	55	17
<code>pagination</code>	48	18
<code>tooltip</code>	40	19
<code>accordion</code>	37	20
<code>offcanvas</code>	27	21
<code>nav-pills</code>	27	22
<code>progress</code>	26	23
<code>toast</code>	17	24
<code>list-group</code>	14	25
<code>nav-tabs</code>	10	26
<code>carousel</code>	4	27
<code>popover</code>	0	—

3.2 Bootstrap JS trigger attributes

Attribute	Occurrences
<code>data-bs-dismiss</code>	567
<code>data-bs-toggle</code> (total)	350
<code>└="modal"</code>	102
<code>└="tab"</code>	81
<code>└="dropdown"</code>	61
<code>└="collapse"</code>	56
<code>└="tooltip"</code>	33

Attribute	Occurrences
<code>└ ="pill"</code>	9
<code>└ ="offcanvas"</code>	7
<code>data-bs-target</code>	189
<code>data-bs-backdrop</code>	66
<code>data-bs-keyboard</code>	31
<code>data-bs-auto-close</code>	21
<code>data-bs-placement</code>	15
<code>data-bs-theme</code>	9
<code>data-bs-parent</code>	4
<code>data-bs-custom-class</code>	4
others (<code>popper</code> , <code>title</code> , <code>ride</code> , <code>pause</code> , <code>original-title</code> , <code>interval</code> , <code>display</code>)	1 each

3.3 Grid

Class family	Occurrences
<code>col-md-*</code>	2,352
<code>row</code>	1,330
<code>container</code>	568
<code>container-fluid</code>	478
<code>g-3</code>	378
<code>col-lg-*</code>	342
<code>col-sm-*</code>	301
<code>col-xl-*</code>	76
<code>gy-*</code>	3
<code>gx-*</code>	0

`col-md-*` outnumbers `col-lg-*` 7:1 — the layout is tuned for tablet-width and simply stretches on desktop. Not a UX4G problem, but a responsive-design debt worth logging.

3.4 Icon systems — four in production

System	Occurrences	Delivery
Bootstrap Icons (<code>bi bi-</code>)	1,281	jsDelivr CDN (1.11.3 in master, 1.10.5 in timetable)
Material Symbols (<code>material-symbols</code>)	1,111	<code>@fontsource-variable</code> , local CSS
Material Icons (<code>material-icons</code>)	1,044	<code>admin_assets/css/material-icons-local.css</code>
Iconify (<code>iconify</code>)	397	Needs Code Inspection — no Iconify runtime found in <code>admin_assets/libs/</code> ; may be dead markup
Font Awesome (<code>fas fa-</code> , <code>fa fa-</code>)	169	Needs Code Inspection — no FA stylesheet found in the load order; these icons may be silently rendering as blank boxes today
Tabler (<code>ti ti-</code>)	9	vestigial

Two of six systems have no confirmed stylesheet. This is a live rendering bug independent of UX4G and should be triaged in Sprint 0.

3.5 Custom CSS

File	Size	!important
<code>admin_assets/css/styles.css</code>	1,002,558 B	5,579
<code>css/custom.css</code>	69,744 B	209

File	Size	!important
css/calendar-admin.css	62,268 B	206
css/sargam-app.css	50,943 B	32
css/notice-memo-discipline.css	45,858 B	—
css/dashboard-main.css	39,944 B	32
admin_assets/css/original/accesibility-style_v1.css	39,903 B	16
admin_assets/css/accesibility-style_v1.css	36,182 B	—
admin_assets/css/plugins/datatables.min.css	34,566 B	18
css/course-repository-admin.css	26,847 B	54
admin_assets/css/sidebar-modern.css	22,225 B	277
css/admin-header.css	22,850 B	195
css/mobile-responsive.css	22,110 B	52
... 20 further files in public/css		
Total	~1.70 MB	7,128

Plus 379 inline `<style>` blocks in 319 Blade files and 2,956 inline `style=""` attributes — neither cacheable nor overridable by a design system.

3.6 Third-party libraries (35 vendored)

@claviska · apexcharts · block-ui · bootstrap · bootstrap-datepicker · bootstrap-switch · bootstrap-tree · datatables.net · datatables.net-bs5 · daterangepicker · dragula · dropzone · fullcalendar · inputmask · jquery-asColor · jquery-asColorPicker · jquery-asGradient · jquery-raty-js · jquery-steps · jquery-ui · jquery-validation · jquery.repeater · jvectormap · magnific-popup · nestable · nouislider-orxe · owl.carousel · prismjs · quill · select2 · simplebar · sweetalert2 · tinymce · typeahead.js · wnumb

3.7 External CDN dependencies (25 hosts)

Host	References	Risk for a government portal
cdn.jsdelivr.net	232	High — third-party, non-Gol, single point of failure
cdn.datatables.net	35	High
fonts.googleapis.com	27	High — GIGW/data-localisation concern
code.jquery.com	19	Critical — jQuery is served entirely off-site; local copy is 0 bytes
cdnjs.cloudflare.com	19	High
fonts.gstatic.com	13	High
upload.wikimedia.org · i.pinimg.com · assets.codepen.io · front.codes	18	Unacceptable — user-content hosts referenced from a government portal
Laravel marketing domains (laravel.com , forge , vapor , nova , envoy , laracasts , laravel-news , herd)	8	Dead scaffolding in welcome.blade.php — delete
Gol hosts (lbsnaa.gov.in , india.gov.in , negd.gov.in , digitalindiaincorporation.in)	20	Acceptable

Recommendation: self-hosting all frontend assets is a prerequisite for GIGW compliance and should be folded into this migration as Sprint 1 work. It is also the mechanism by which UX4G's missing fonts/sprites get fixed (\$1.5 R-3).

4. UX4G Mapping Matrix

4.1 What UX4G v2.0.8 actually ships

Category	Contents
Bootstrap-parity components	Accordion, Alerts, Badge, Breadcrumb, Button Group, Buttons, Card, Carousel, Close Button, Collapse, Dropdowns, List Groups, Modal, Navbar, Navs & Tabs, Offcanvas, Pagination, Placeholders, Popovers, Progress, Scrollspy, Spinners, Toasts, Tooltips
UX4G-additional	Chips (<code>.chip</code> <code>.chips</code> <code>.chips-input-wrapper</code> <code>.chips-placeholder</code> <code>.chips-transition</code> <code>.chip-opacity</code>), Stepper (25 classes incl. <code>.stepper-vertical</code> <code>.stepper-mobile</code> <code>.stepper-head-check</code> <code>.stepper-progress-bar</code> <code>.stepper-invalid</code>), Search, Date & Time Picker
Grid	Columns, Containers, CSS Grid, Gutters, Breakpoints, Z-index — Bootstrap-identical
Forms	Form Control, Floating Labels, Checks & Radios, Input Group, Select, Range, Validation
Utilities	Background, Borders, Colors, Display, Flex, Spacing, Text, Sizing, Shadows
Helpers	Color & Background, Position, Stacks, Text Truncation, Visually Hidden
Dependency	Popper.js only. No Bootstrap. No jQuery.

4.2 The mapping matrix

Legend — **Complexity**: Trivial = class contract identical, only visual reconciliation; Low ≤ 1 d; Medium 1–3 d; High > 3 d.

#	Current (Bootstrap 5.3)	UX4G equivalent	Complexity	Breaking changes	Dependencies	A11y improvement	Effort
1	Card ×2,288	Card	Trivial	None — classes identical	None	Neutral	4 h (visual reconcile)
2	Buttons ×2,202	Buttons + <code>.ripple-button</code>	Trivial	None. Tonal variants Needs Code Inspection	None	Focus ring meets 2.4.7	8 h
3	Form Control ×1,697	Form Control	Low	Height/padding differ → 40 dense forms need reflow QA	None	Better contrast on <code>:disabled</code>	24 h
4	Table ×997	(no UX4G table component) — Bootstrap <code>.table-*</code> present in <code>ux4g-min.css</code>	Low	None	DataTables theme	Needs manual <code><caption></code> / <code>scope</code>	16 h
5	Form Select ×861	Select	Low	Select2 re-skin needed (\$8)	select2	Neutral	16 h
6	Form Check ×811	Checks & Radios	Trivial	None	None	Larger hit target	6 h
7	Badge ×600	Badge	Trivial	Palette shift only	None	Neutral	3 h
8	Alert ×384	Alerts	Trivial	None	None	Add <code>role="alert"</code> (54 sites lack it)	6 h
9	Modal ×207	Modal	Trivial	<code>data-bs-*</code> + <code>.bs.modal</code> events identical; JS global stays <code>bootstrap</code>	Popper	Focus trap already correct	8 h
10	Breadcrumb ×135	Breadcrumb	Low	Custom <code>breadcrumb.css</code> (8.2 KB) must be retired	<code>components/breadcrum.blade.php</code>	<code>aria-current="page"</code>	8 h
11	Spinners ×95	Spinners	Trivial	None	None	Add <code>role="status"</code>	2 h
12	Collapse ×94	Collapse	Trivial	None	None	<code>aria-expanded</code> present ×210	4 h
13	Input Group ×92	Input Group	Trivial	None	None	Neutral	4 h
14	Dropdown ×59	Dropdowns	Trivial	None	Popper	Keyboard nav parity	4 h
15	Navbar ×55	Navbar	Medium	Custom header (<code>admin-header.css</code> , 22.9 KB / 195 <code>!important</code>) must be rebuilt on UX4G primitives	RBAC resolver	Landmark roles	32 h
16	Pagination ×48	Pagination	Low	Laravel paginator view must be republished	Laravel pagination views	<code>aria-label</code> on links	8 h
17	Tooltip ×40	Tooltips	Trivial	None	Popper	Tooltips are not keyboard-reachable today — fix during migration	6 h
18	Accordion ×37	Accordion	Trivial	None	None	Neutral	3 h
19	Offcanvas ×27	Offcanvas	Trivial	None	None	Focus trap	3 h
20	Nav Pills ×27 / Tabs ×10	Navs & Tabs	Trivial	None; but the 5-pane RBAC tab system (\$2.2) is custom and frozen	Sidebar resolver	<code>role="tablist"</code> audit	8 h
21	Progress ×26	Progress	Trivial	None	None	<code>aria-valuenow</code> audit	2 h
22	Toast ×17	Toasts	Low	Global success toast is SweetAlert2-based, not BS Toast — decide whether to converge	sweetalert2	<code>aria-live</code>	8 h

#	Current (Bootstrap 5.3)	UX4G equivalent	Complexity	Breaking changes	Dependencies	A11y improvement	Effort
23	List Group ×14	List Groups	Trivial	None	None	Neutral	2 h
24	Carousel ×4	Carousel	Trivial	None	None	Pause control needed	2 h
25	Popover ×0	Popovers	N/A	Not used	—	—	0 h
26	(none)	Chips — new	Low	Net-new capability; adopt for filter pills	None	Removable-chip labelling	12 h
27	<code>components/estate-workflow-stepper</code> + <code>partials/step-indicator</code> + <code>jquery-steps</code>	Stepper — new	Medium	Replaces a jQuery plugin with CSS/JS-lite	Removes <code>jquery-steps</code>	Progress semantics	24 h
28	Bootstrap Icons + Material Symbols + Material Icons + Iconify + FA	UX4G gov icon sprites + one retained icon font	High	4,011 icon references; sprites 404 on CDN (\$1.5 R-3)	Self-hosting	Decorative icons need <code>aria-hidden</code>	64 h
29	<code>daterangepicker</code> + <code>bootstrap-datepicker</code>	UX4G Date & Time Picker	Medium	Needs Code Inspection — no <code>.datepicker</code> / <code>.timepicker</code> classes found in <code>ux4g-min.css</code> ; component may be JS-only or v3-only. Do not plan on this until verified.	—	—	TBD
30	Custom search boxes / <code>dropdown-search.js</code>	Search	Low	6 <code>.search-*</code> rules found in CSS	None	<code>role="search"</code>	12 h

4.3 Why "Trivial" is honest here

For rows marked Trivial the required Blade change is **zero**. `<div class="card">` is a valid UX4G card. `data-bs-toggle="modal"` is the documented UX4G API. `new bootstrap.Modal( el )` is the working constructor. The only work is confirming the new visual defaults don't break a layout — which is QA time, not development time, and is priced in \$13.

5. Component-wise Migration Plan

Component	Current usage	UX4G alternative	Files affected	Dependencies	Order	Hours	Risk	Regression risk	Testing	Rollback
Foundation / tokens	--bs-primary:#004384 in styles.css	--bs-primary:#613AF5 (or approved override)	pre_header.blade.php , new ux4g-tokens.css	R-1 decision	1	40	High	App-wide	Full visual regression baseline	Remove one <link>
Self-hosting + fonts	25 CDN hosts; jQuery off-site	Local public/vendor/ux4g/ incl. Noto Sans + sprites	pre_header , timetable , fc/pre_header	R-3	2	48	Medium	Asset 404s	Network-panel audit per layout	Revert <link> / <script>
Typography	Mixed; no single family token	Noto Sans via --bs-body-font-family	ux4g-tokens.css	Fonts self-hosted	3	24	Medium	Line-height shifts in dense tables	Screenshot diff on 20 densest pages	Token revert
Spacing	spacing-system.css (4.8 KB, 42 !important)	UX4G spacing utilities	spacing-system.css → delete	Tokens	4	24	Medium	Layout drift	Grid-heavy page sweep	Restore file
Buttons	2,202	Buttons	0 Blade edits; theme only	Tokens	5	8	Low	Low	Visual	Token revert
Forms / Inputs	1,697 + 861 + 811	Form Control / Select / Checks	~40 dense forms need reflow	Buttons	6	24	Medium	Medium	Form-submit E2E per module	Per-page CSS shim
Validation	jquery-validation ×12 + custom	UX4G Validation classes	fc-form-validation.css	Forms	7	24	Medium	Submit-blocking	Every validated form manually	Keep plugin
Cards	2,288	Card	Theme only	Tokens	8	4	Low	Low	Visual	Token revert
Tables	997 + 101 DataTable inits	.table-* + DT bs5 theme	datatable-global-ui.js , datatable.min.css	Forms	9	32	High	High — see §8	All 101 DT pages, both modes	Restore DT theme CSS
Navbar / Header	admin-header.css 195 !important	Navbar	header.blade.php , admin-header.css	Tokens	10	32	High	Global	Every page renders header	Restore CSS
Sidebar	sidebar-menu-enhanced.css + sidebar-modern.css (277 !important) + 5 JS files	Offcanvas + List Group	admin/layouts/sidebar* , 4 JS files	RBAC resolver — frozen	11	48	Critical	Global	Full RBAC matrix per role	Restore CSS+JS
Breadcrumb	135	Breadcrumb	components/breadcrum.blade.php , breadcrumb.css	Tokens	12	8	Low	Low	Spot-check	Restore CSS
Dropdown	59	Dropdowns	Theme only	Popper	13	4	Low	Low	Visual	Token revert
Modal	207	Modal	Theme only	Popper	14	8	Low	Low	Open/close/submit per modal	Token revert
Offcanvas	27	Offcanvas	Theme only	—	15	3	Low	Low	Visual	—
Accordion	37	Accordion	Theme only	—	16	3	Low	Low	Visual	—
Tabs / Pills	37	Navs & Tabs	Theme only — RBAC pane logic frozen	Sidebar	17	8	Medium	Medium	Tab-switch per module	—
Pagination	48	Pagination	Laravel paginator views	Tokens	18	8	Low	Low	Paginated list sweep	Republish old view
Alerts	384	Alerts	Theme only	Tokens	19	6	Low	Low	Visual	—
Badges	600	Badge	Theme only	Tokens	20	3	Low	Low	Visual	—
Toast	17 + SweetAlert2 global toast	Toasts	sargam-success-toast.js	Decision	21	8	Medium	Medium	Success-path E2E	Keep SweetAlert2
Tooltip	40	Tooltips	Theme + init	Popper	22	6	Low	Low	Hover + keyboard	—
Popover	0	Popovers	—	—	23	0	—	—	—	—
Carousel	4	Carousel	Theme only	—	24	2	Low	Low	Visual	—
Stepper	jquery-steps + 2 custom Blades	Stepper	estate-workflow-stepper , step-indicator	Forms	25	24	Medium	Medium	Estate + FC wizards E2E	Keep jquery-steps
Chips	(none)	Chips	datatable-global-ui.js filter pills	Tables	26	12	Low	Low	Filter behaviour	Revert
Search	dropdown-search.js	Search	dropdown-search.js	—	27	12	Low	Low	Search UX	Revert
Date Picker	daterangepicker ×72	Unverified — see §4.2 row 29	—	R-2	28	TBD	High	High	—	Keep plugin
Charts	ApexCharts (516 KB) — 1 Blade	(none in UX4G)	dashboard_statistics/charts.blade.php	Tokens	29	8	Low	Low	Chart render	—
Icons	4,011 refs / 6 systems	Consolidate to 1 + gov sprites	~400 Blades	Self-hosting	30	64	High	High	Visual sweep	Keep all fonts loaded
Custom widgets	Calendar (62 KB CSS), timetable, ID cards	Retain, re-tokenise	per §17	Tokens	31	80	Medium	Medium	Per-widget	Per-file
Page sweep	494 admin + 31 FC + 12 other	—	537 views	All above	32	~400	Medium	Medium	Per-page UAT	Per-page

6. Migration Sequence

The sequence is ordered so that **every stage leaves a shippable application**, and so that high-blast-radius work happens while the regression suite is freshest.

STAGE 0 – STABILISE (no UX4G yet)

Unify Bootstrap to one version (5.3.2 / 5.3.3 / 5.3.6 → one)

Restore local jQuery (public/js/jquery-3.7.1.min.js is 0 bytes)

Triage Iconify (397) + Font Awesome (169) – confirm or delete

Delete Laravel marketing CDN refs in welcome.blade.php

Establish Playwright visual-regression baseline over top 60 pages

↓

STAGE 1 – FOUNDATION

Self-host UX4G 2.0.8 (CSS, JS, Noto Sans, gov icon sprites)

Insert ux4g-min.css at load position 0, behind existing theme

Create public/css/ux4g-tokens.css (single token override file)

Decide R-1 (brand colour) and encode it here

Verify: zero visual change expected at this point

↓

STAGE 2 – TOKENS

Typography → Colors → Radius → Spacing → Elevation → Focus/Hover/Disabled

↓

STAGE 3 – ATOMS

Buttons → Badges → Alerts → Spinners → Progress

↓

STAGE 4 – FORMS

Form Control → Select → Checks/Radios → Input Group → Floating Labels → Validation

↓

STAGE 5 – SURFACES

Cards → List Groups → Tables

↓

STAGE 6 – CHROME

Navbar/Header → Sidebar → Breadcrumb → Tabs/Pills ← highest risk; RBAC frozen

↓

STAGE 7 – OVERLAYS

Dropdown → Modal → Offcanvas → Tooltip → Toast → Accordion → Carousel

↓

STAGE 8 – NEW UX4G

Chips → Stepper → Search → Date Picker (if R-2 resolves)

↓

STAGE 9 – PLUGINS

DataTables → Select2 → Summernote → FullCalendar → SweetAlert2 → Dropzone

↓

STAGE 10 – WIDGETS

Calendar → Timetable → Dashboard widgets → ID cards → Course repository

↓

STAGE 11 – PAGES

494 admin + 31 FC + 12 other, module by module

↓

STAGE 12 – CLEANUP

Delete dead CSS · purge !important · remove inline styles · ally remediation · perf pass

Rationale for the ordering. Tokens first because in a Bootstrap-compatible system a token change propagates to *every* component for free — doing components first would mean re-doing them. Chrome (Stage 6) before plugins (Stage 9) because DataTables and Select2 render *inside* page chrome and would need re-testing anyway. Pages last because per-page fixes are only valid once the shared layer is stable.

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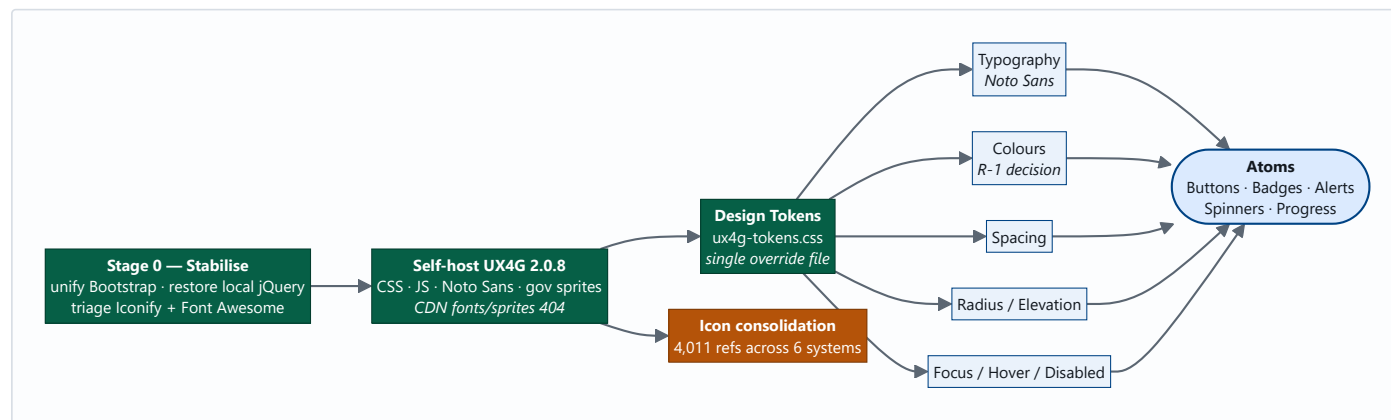
Page 15 of 34

7. Dependency Graph

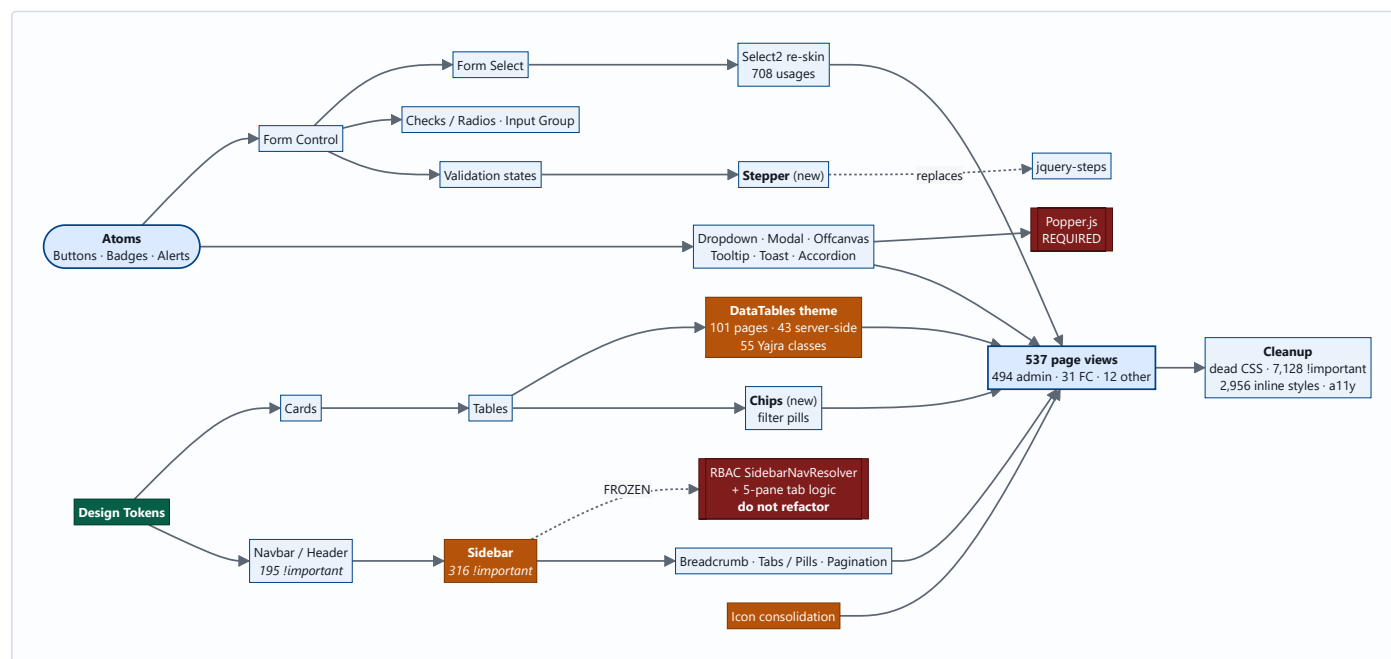
Split into two views for legibility. **7.1** covers the shared foundation every component inherits; **7.2** covers the component-to-page dependencies that determine sprint ordering.

Colour key — ■ green = foundation (must land first) · ■ amber = high-risk, dedicated sprint · ■ red = frozen contract or hard external dependency.

7.1 Foundation layer



7.2 Component and page dependencies



Critical path: Stage 0 → Self-host → Tokens → Navbar → Sidebar → Pages. The sidebar is the longest-pole item because it couples to the frozen RBAC resolver.

8. jQuery Plugin Analysis

`$(...)` appears **5,851 times across 192 Blade files**. jQuery is not removable in this migration and should not be attempted.

Plugin	Usage (Blade)	UX4G replacement?	Verdict	Rationale	Risk
DataTables (+ <code>datatables.net-bs5</code>)	1,587 refs · 101 pages · 43 <code>serverSide</code> · 55 Yajra PHP classes	No	KEEP	Yajra server-side pagination is backend-coupled; replacing it violates the no-backend-change constraint. Re-theme only.	High — <code>dataTables.bootstrap5</code> targets <code>.page-link</code> , <code>.form-select</code> , <code>.table</code> ; UX4G ships all three, so the theme <i>should</i> hold, but must be verified per page. Never convert a Yajra table to custom-DOM/colVis — it breaks init.
Select2 (theme <code>bootstrap-5</code>)	708	Partial — UX4G <code>Select</code> covers native only	KEEP + re-skin	Multi-select, AJAX search and tagging have no UX4G equivalent	Medium — <code>select2-theme.css</code> (5.6 KB) + <code>choices-theme.css</code> (3.1 KB) both exist; consolidate to one
SweetAlert2	285 (<code>Swal</code>) + 25	Partial — UX4G Modal/Toast	KEEP	285 call sites; the global success toast is a documented project pattern. Converging is out of scope.	Low
SimpleBar	257	No	KEEP	Custom scrollbars in sidebar	Low
daterangepicker	72	Unverified — UX4G lists a Date & Time Picker but no <code>.datepicker</code> / <code>.timepicker</code> classes exist in <code>ux4g-min.css</code>	KEEP pending R-2	Do not schedule removal until the component is confirmed to exist in a self-hostable build	High
Summernote 0.8.18	67	No	KEEP, but self-host	Loaded from <code>cdn.jsdelivr.net</code> , not vendored — a CDN outage breaks every rich-text field	Medium
FullCalendar	74	No	KEEP	No UX4G calendar. 288 KB. Re-tokenise <code>calendar-admin.css</code> (62 KB) only.	Medium
Dropzone	20	No	KEEP	No UX4G file-upload component	Low
jquery-validation	12	Partial — UX4G Validation is CSS-state only	KEEP	UX4G provides <code>.is-valid</code> / <code>.is-invalid</code> styling, not validation logic	Medium
ApexCharts (516 KB)	3 refs · 1 Blade	No	REVIEW — likely remove	516 KB shipped for one file (<code>dashboard_statistics/charts.blade.php</code>). Load it lazily on that route only.	Low
jquery-steps	2	Yes — UX4G Stepper	REPLACE	Genuine 1:1 replacement with better a11y	Medium
TinyMCE (428 KB)	0	—	DELETE	Zero references in Blade	None
Quill (216 KB)	0	—	DELETE	Zero references	None
jQuery UI (255 KB)	0 direct	—	AUDIT then delete	Needs Code Inspection — may be a transitive dep of another plugin	Low
jvectormap (169 KB incl. US map)	0	—	DELETE	A US county map on an Indian government portal	None
owl.carousel (44 KB)	0	UX4G Carousel	DELETE	Zero references	None
inputmask (104 KB)	0	No	DELETE	Zero references	None
nouislider + <code>wnumb</code>	0	UX4G Range	DELETE	Zero references	None
typeahead.js (27 KB)	0	UX4G Search	DELETE	Zero references	None
magnific-popup	0	UX4G Modal	DELETE	Zero references	None
prismjs (60 KB)	0	—	DELETE	Syntax highlighting has no place in an ERP	None
<i>nestable</i> / <i>dragula</i> / <i>bootstrap-tree</i> / <i>bootstrap-switch</i> / <i>block-ui</i> / <i>jquery-ratyjs</i> / <i>jquery-asColor</i> / <i>@claviska</i> / <i>jquery.repeater*</i>	0–1	—	AUDIT then delete	<code>repeater</code> has 1 ref — verify before removing	Low

8.1 Estimated dead weight

Removing the zero-reference libraries reclaims approximately **1.4 MB** of shipped JavaScript (TinyMCE 428 KB + Quill 216 KB + jQuery UI 255 KB + jvectormap 169 KB + owl 44 KB + inputmask 104 KB + prism 60 KB + nouislider 24 KB + typeahead 27 KB + asColorPicker 34 KB + others). Fold this into Stage 12.

9. CSS Refactoring Plan

9.1 The problem, quantified

Symptom	Measure
Total shipped custom/theme CSS	~1.70 MB
<code>!important</code> declarations	7,128
↳ in <code>styles.css</code> alone	5,579 (78 %)
Inline <code><style></code> blocks	379 in 319 files
Inline <code>style="..."</code> attributes	2,956
Stylesheets in the critical path	14
Duplicate a11y stylesheets	<code>accessibility-style_v1.css</code> exists in both <code>css/</code> (36 KB) and <code>css/original/</code> (40 KB)

Why this matters for UX4G specifically: 7,128 `!important` rules were written to beat the *current* theme's specificity. They will beat UX4G's tokens with exactly the same force. **Every one of them is a place where the design system will silently fail to apply.** This — not component renaming — is the real migration cost.

9.2 Target architecture

public/css/	
├── ux4g/	← self-hosted, NEVER edited
│ ├── ux4g-min.css	(269 KB, v2.0.8, pinned)
│ ├── fonts/NotoSans-*.woff2	(fixes the CDN 404)
│ └── img/*-icons.svg	(fixes the sprite 404)
├── ux4g-tokens.css	← ONLY file that overrides --bs-* tokens
│ (brand colour, radius, elevation, focus)	
└── sargam-app.css	← ONE consolidated custom stylesheet
├── @layer base	: resets, typography that UX4G doesn't cover
├── @layer components	: .sg-* components (BEM-ish, no !important)
├── @layer widgets	: calendar, timetable, ID card, course repo
└── @layer utilities	: project-specific utilities only

Everything else in `public/css/` and `public/admin_assets/css/` is **deleted or absorbed**.

This aligns with the existing project convention that `public/css/sargam-app.css` is the single consolidated custom stylesheet. The migration formalises that rule rather than inventing a new one.

9.3 Refactor rules

- Cascade layers over `!important` — with the ordering corrected.** The naive `@layer ux4g, app` is **wrong here**: `ux4g-min.css` ships **1,668 `!important` declarations of its own**, and the CSS cascade *reverses* layer order for `!important` (an important declaration in an earlier layer beats one in a later layer). Layering UX4G first would make its 1,668 important rules beat everything the project writes. **Preferred fix:** strip `!important` from the self-hosted UX4G build at vendor time — possible precisely because R-3 mandates self-hosting — then `@layer ux4g, app` behaves as intended. **Fallback if the build step is rejected:** declare `@layer app-important, ux4g, app;` — project `!important` rules go in `app-important` (first, so they win the reversed important order), project normal rules in `app` (last, so they win the normal order).
- Token-first.** No hard-coded hex/px in `sargam-app.css`. If a value is needed twice, it is a token.
- Naming.** `.sg-<block>__<element>--<modifier>`. Do not invent a parallel `.ux4g-*` namespace — v2.0.8 does not use one.
- Inline style budget.** Cap at 0 new inline `style=` attributes; burn down the 2,956 existing ones opportunistically during the Stage 11 page sweep (not as dedicated work).
- Inline `<style>` blocks.** All 379 must be either promoted to `sargam-app.css` or scoped under a page class. Note the known hazard: Blade directives inside HTML comments still execute and, combined with `CompressResponse`, can produce `ERR_CONTENT_DECODING_FAILED` blank pages — escape with `{{--@@section --}}`.
- Dead-CSS detection.** Run PurgeCSS in *report-only* mode against all 853 Blade files plus the JS that injects classes at runtime (`datatable-global-ui.js`, `custom.js`). **Do not auto-purge** — DataTables and Select2 generate class names at runtime and will be false-positived.

9.4 Sequenced CSS work

Step	Action	Files	Hours
C1	Introduce <code>@layer</code> , wrap UX4G + app	<code>pre_header.blade.php</code>	8

Step	Action	Files	Hours
C2	Create <code>ux4g-tokens.css</code> , encode R-1	new	16
C3	Delete <code>spacing-system.css</code> , map to UX4G utilities	1 file, 42 <code>!important</code>	16
C4	Absorb <code>breadcrumb.css</code> into <code>sargam-app.css</code>	1 file	8
C5	Rebuild header on UX4G Navbar; delete <code>admin-header.css</code>	1 file, 195 <code>!important</code>	32
C6	Merge <code>sidebar-menu-enhanced.css</code> + <code>sidebar-modern.css</code>	2 files, 316 <code>!important</code>	40
C7	De-duplicate <code>accessibility-style_v1.css</code> (2 copies)	2 files	8
C8	Absorb 20 per-module stylesheets in <code>public/css</code>	20 files	64
C9	Reduce <code>styles.css</code> to a UX4G-compatible residue	1 MB, 5,579 <code>!important</code>	120
C10	Promote 379 inline <code><style></code> blocks	319 files	64
C11	PurgeCSS report + manual dead-code removal	all	32
	Subtotal		408

10. Accessibility Gap Analysis

10.1 Baseline (static counts, `resources/views/**`)

Marker	Count	Reading
<code><label</code>	2,547	Good — vs 3,369 form controls, so ~76 % nominal coverage
<code>aria-label</code>	1,285	Good
<code>aria-hidden</code>	1,074	Good — mostly decorative icons
<code>role=</code>	1,017	Good
<code>aria-labelledby</code>	324	Adequate
<code>tabindex</code>	300	Audit — positive values are a 2.4.3 violation
<code>alt=</code>	268	Audit — must confirm none are empty-on-informative
<code>aria-expanded</code>	210	Good — matches 56 collapse + 61 dropdown + sidebar
<code>visually-hidden</code>	117	Good (Bootstrap 5 idiom)
<code>lang=</code>	99	Good
<code>aria-describedby</code>	54	Low — 384 alerts and ~3,369 inputs; error messages likely unassociated
<code>aria-live</code>	31	Low — 101 DataTables pages update async with no announcement
<code>skip-link</code>	9	Critical gap — 537 pages, 9 skip links
<code>sr-only</code>	3	Bootstrap 4 legacy → migrate to <code>visually-hidden</code>

10.2 Gap report — WCAG 2.1 AA + GIGW 3.0

ID	Criterion	Standard	Severity	Finding	Remediation	Hours
A-1	2.4.1 Bypass Blocks	AA / GIGW	Critical	9 skip-links for 537 pages. <code>fc.layouts.master</code> does it correctly (<code><main id="content" tabindex="-1"></code>); <code>admin.layouts.master</code> does not.	Add skip-link + <code><main></code> landmark to <code>admin.layouts.master</code> → fixes 494 pages in one edit	8
A-2	1.4.3 Contrast	AA	Critical	Measured. UX4G primary <code>#613AF5</code> on white = 6.12:1 (AA); current <code>#004384</code> = 9.83:1 (AAA). But three shipped UX4G button variants fail AA: <code>.btn-info</code> 2.56:1, <code>.btn-success</code> 3.73:1, <code>.btn-warning</code> 3.86:1 — all pair <code>#fff</code> text with a too-light background. Affects 248 buttons + 325 <code>bg-*</code> + 248 <code>text-*</code> usages. See the Decision Note (R-1).	Override the three failing variants in <code>ux4g-tokens.css</code> ; then automated contrast sweep	32
A-3	4.1.3 Status Messages	AA	High	31 <code>aria-live</code> vs 101 async DataTables + 285 SweetAlert calls	Add <code>aria-live="polite"</code> to DT wrappers and the global toast	16
A-4	3.3.1 / 3.3.3 Error ID	AA / GIGW	High	54 <code>aria-describedby</code> across ~3,369 inputs — validation errors are almost certainly visual-only	Wire <code>.invalid-feedback</code> to inputs via <code>aria-describedby</code> in the shared form components	32
A-5	2.1.1 Keyboard	A	High	33 <code>data-bs-toggle="tooltip"</code> on non-focusable elements — tooltip content is mouse-only	Move tooltips to <code><button> / tabindex="0"</code>	12
A-6	2.4.3 Focus Order	A	Medium	300 <code>tabindex</code> — positive values break document order	Audit; allow only <code>0</code> / <code>-1</code>	16
A-7	2.4.7 Focus Visible	AA	Medium	7,128 <code>!important</code> rules very likely clobber <code>:focus-visible</code>	Define focus ring as a token in <code>ux4g-tokens.css</code> , layer it above app CSS	16
A-8	1.1.1 Non-text	A	Medium	268 <code>alt=</code> for 268+ images; 4,011 icon refs need <code>aria-hidden</code> on decorative ones	Sweep during icon consolidation	24
A-9	1.3.1 Info & Relationships	A	Medium	997 tables — <code><caption></code> , <code>scope</code> , <code><th></code> usage unverified	Table semantics sweep	32
A-10	1.4.4 Resize / 1.4.10 Reflow	AA	Medium	2,956 inline <code>style=</code> (many with fixed <code>px</code>) resist zoom	Burn down during page sweep	—
A-11	GIGW 3.0 — a11y widget	GIGW	Medium	<code>accessibility-style_v1.css</code> present but loaded on only 6 of 537 views	Promote to <code>admin.layouts.master</code>	8
A-12	1.4.12 Text Spacing	AA	Low	Fixed line-heights in dense tables	Token-driven	8
A-13	4.1.2 Name/Role/Value	A	Low	Custom 5-pane tab system — <code>role="tablist"</code> / <code>aria-selected</code> unverified	Needs Code Inspection	8

ID	Criterion	Standard	Severity	Finding	Remediation	Hours
A-14	Legacy	—	Low	3 <code>sr-only</code> (Bootstrap 4)	Replace with <code>visually-hidden</code>	1
					Subtotal	213

10.3 The accessibility upside

This is the strongest business case for the migration. UX4G ships WCAG 2.1 AA-conformant component defaults, so items A-7, A-12 and much of A-2 are fixed **by adopting the tokens**, not by writing remediation code. A-1 is a single edit to one layout covering 494 pages. The genuinely manual work — A-4, A-8, A-9 — is ~88 h out of 213 h.

11. Design Token Mapping

11.1 Colour

Token	Current (<code>styles.css</code>)	UX4G 2.0.8	Δ	Action
<code>--bs-primary</code>	<code>#004384</code> (LBSNAA navy)	<code>#613AF5</code> (violet)	Total	R-1 decision
<code>--bs-blue</code>	—	<code>#613AF5</code>	—	adopt
<code>--bs-secondary</code>	—	<code>#938BB6</code>	—	adopt
<code>--bs-success</code>	—	<code>#3C9718</code>	—	adopt
<code>--bs-info</code>	—	<code>#00AAFF</code>	—	adopt
<code>--bs-warning</code>	—	<code>#B77224</code>	—	adopt
<code>--bs-danger</code>	—	<code>#B7131A</code>	—	adopt
<code>--bs-light</code>	—	<code>#f8f9fa</code>	none	adopt
<code>--bs-dark</code>	—	<code>#212121</code>	—	adopt
<code>--bs-red / orange / yellow / green / teal / cyan</code>	—	<code>#B7131A</code> <code>#fd7e14</code> <code>#FFC53F</code> <code>#3C9718</code> <code>#20c997</code> <code>#00AAFF</code>	—	adopt
<code>--bs-gray-{50...900}</code>	Bootstrap default	<code>#F3F3F3</code> <code>#DDD</code> <code>#C6C6C6</code> <code>#B0B0B0</code> <code>#9B9B9B</code> <code>#868686</code> <code>#727272</code> <code>#5E5E5E</code> <code>#4B4B4B</code> <code>#393939</code>	Different scale — UX4G adds <code>gray-50</code>	Map carefully; 10 steps not 9
<code>--bs-body-color</code>	—	<code>#212121</code>	—	adopt
<code>--bs-body-bg</code>	—	<code>#fff</code>	—	adopt
<code>--bs-link-color</code>	—	<code>#613AF5</code>	R-1	follow primary
<code>--bs-link-hover-color</code>	—	<code>#774BFF</code>	R-1	follow primary
<code>--bs-border-color</code>	—	<code>#dee2e6</code>	none	adopt
<code>--bs-highlight-bg</code>	—	<code>#fff3cd</code>	—	adopt
<code>--light-purple</code>	—	147, 139, 182	UX4G-only, non- <code>--bs-</code>	note the naming inconsistency

On R-1. UX4G violet measures 6.12:1 on white (AA); LBSNAA navy measures 9.83:1 (AAA). Retaining navy is therefore *more* accessible, not a compliance compromise.

Critical implementation caveat. `var(--bs-primary)` appears **exactly once** in the entire 269 KB `ux4g-min.css`, while the literal `#613AF5` appears **38 times, 5 of them with `!important`** (e.g. `.btn-primary{--bs-btn-bg:#613AF5!important}`). **Overriding the `--bs-primary` token therefore does almost nothing** — UX4G hard-codes its brand colour rather than deriving it. Re-branding requires a compiled override layer, not a variable change. Full analysis and remediation in the Decision Note (R-1).

11.2 Typography

Token	UX4G value	Note
<code>--bs-body-font-family</code>	'Noto Sans', system-ui, -apple-system, 'Segoe UI', 'Noto', 'Helvetica', 'Arial', sans-serif, ...	Noto Sans is essential — it is the GoI standard and the only family with full Devanagari + 22-script coverage
<code>--bs-font-sans-serif</code>	"Noto Sans", sans-serif	
<code>--bs-font-monospace</code>	'Noto Sans'	Defect in UX4G — the monospace token is not a monospace font. Override locally.
<code>--bs-body-font-size</code>	1rem	
<code>--bs-body-font-weight</code>	400	
<code>--bs-body-line-height</code>	1.5	

Blocker: `@font-face` in `ux4g-min.css` points to `../fonts/NotoSans-Regular.{woff2,woff,ttf}`; all three **404 on the CDN**. Source Noto Sans from Google Fonts / the UX4G GitHub and self-host at the matching relative path.

11.3 Radius

Token	UX4G	Bootstrap default	Δ
<code>--bs-border-radius</code>	<code>0.375rem</code>	<code>0.375rem</code>	none
<code>--bs-border-radius-sm</code>	<code>0.25rem</code>	<code>0.25rem</code>	none
<code>--bs-border-radius-lg</code>	<code>0.5rem</code>	<code>0.5rem</code>	none
<code>--bs-border-radius-xl</code>	<code>1rem</code>	<code>1rem</code>	none
<code>--bs-border-radius-2xl</code>	<code>2rem</code>	<code>2rem</code>	none
<code>--bs-border-radius-pill</code>	<code>6.25rem</code>	<code>50rem</code>	differs — cosmetic only

11.4 Spacing, breakpoints, borders

Aspect	Finding
Spacing	84 distinct <code>.{m,p}{t,b,s,e,x,y}-{0..5}</code> utilities — Bootstrap's <code>0 / .25 / .5 / 1 / 1.5 / 3 rem</code> scale, unchanged
Breakpoints	<code>576 / 768 / 992 / 1200 / 1400 px</code> — identical to Bootstrap 5
Grid	60 <code>.col-{bp}-{n}</code> classes — identical
Border	<code>--bs-border-width: 1px</code> , <code>--bs-border-style: solid</code> , <code>--bs-border-color-translucent: rgba(0,0,0,.175)</code> — identical

This is why the grid migration cost is effectively zero. All 2,352 `col-md-*`, 1,330 `row` and 478 `container-fluid` usages are already valid UX4G.

11.5 Elevation, states, icons

Aspect	Status
Elevation / shadows	<code>.shadow</code> utilities present (7 rules). No dedicated <code>--bs-box-shadow-*</code> token block observed in <code>:root</code> . Needs Code Inspection — define project elevation tokens in <code>ux4g-tokens.css</code> .
Focus	No <code>--bs-focus-ring-*</code> in the <code>:root</code> block inspected. Needs Code Inspection ; define explicitly (ties to A-7).
Hover	<code>--bs-link-hover-color: #774BFF</code> only. Component hovers are hard-coded in rules, not tokenised.
Disabled	No disabled token. Handled per-component.
Icons	5 sprite sheets referenced: <code>common-gov-icons.svg</code> , <code>country-icons.svg</code> , <code>social-icons.svg</code> , <code>state-icons.svg</code> , <code>ut-icons.svg</code> — all 404 on the CDN . These are genuinely valuable (state/UT icons for a government portal) and are a strong reason to obtain the full distribution from NeGD.

12. Sprint-wise Roadmap

Two-week sprints. Team: **FE1, FE2, FE3** (frontend), **QA1, DES** (designer, 0.4 FTE).

Sprint	Focus	Components	Priority	Depends on	SP	Hours	Risk	Owner	Status	Testing
S0	Stabilise & baseline	BS version unification, jQuery restore, icon triage, Playwright baseline (60 pages)	P0	—	21	84	Med	FE1, QA1	Not started	Visual baseline capture
S1	Foundation	Self-host UX4G 2.0.8 + Noto Sans + sprites; @layer architecture; ux4g-tokens.css	P0	S0, R-1, R-2, R-3	26	104	High	FE1, DES	Not started	Zero-visual-diff assertion
S2	Tokens	Typography, colours, radius, spacing, elevation, focus	P0	S1	21	112	High	FE1, DES	Not started	Full 60-page visual diff
S3	Atoms	Buttons, badges, alerts, spinners, progress	P1	S2	13	52	Low	FE2	Not started	Component gallery
S4	Forms I	Form control, select, checks/radios, input group, floating labels	P0	S3	21	88	Med	FE2, FE3	Not started	Form-submit E2E x40
S5	Forms II + Select2	Validation states, Select2 re-skin, choices-theme consolidation	P0	S4	18	72	Med	FE2	Not started	Validation E2E
S6	Surfaces	Cards, list groups, tables, styles.css reduction begins	P1	S3	21	84	Med	FE3	Not started	Visual
S7	DataTables	DT bs5 theme reconciliation, datatable-global-ui.js, Chips filter pills	P0	S6	26	104	High	FE3, QA1	Not started	All 101 DT pages, both modes
S8	Chrome I — Header	Navbar rebuild, delete admin-header.css	P0	S2	21	84	High	FE1	Not started	Every page renders header
S9	Chrome II — Sidebar	Sidebar CSS merge (316 !important), RBAC contract preserved	P0	S8	34	136	Critical	FE1, QA1	Not started	Full RBAC x role matrix
S10	Chrome III	Breadcrumb, tabs/pills, pagination	P1	S9	16	64	Med	FE2	Not started	Nav E2E
S11	Overlays	Dropdown, modal, offcanvas, tooltip, toast, accordion, carousel	P1	S2	18	72	Med	FE2	Not started	Open/close/submit x207 modals
S12	New UX4G components	Stepper (replaces jquery-steps), Search, Chips rollout	P2	S5	18	72	Med	FE3	Not started	Wizard E2E
S13	Plugins & purge	Summernote self-host, FullCalendar re-tokenise, SweetAlert2, delete 1.4 MB dead libs	P1	S11	21	84	Med	FE3	Not started	Plugin smoke
S14	Accessibility	A-1...A-14 remediation	P0	S9	26	112	Med	FE2, QA1	Not started	axe-core + NVDA + keyboard
S15	Widgets	Calendar, timetable, dashboard, ID cards, course repository	P2	S7	21	84	Med	FE3	Not started	Per-widget
S16–S20	Page sweep	494 admin + 31 FC pages, module by module (see §13.2)	P1	S1–S15	100	400	Med	FE1-3, QA1	Not started	Per-page UAT
S21	CSS cleanup	!important purge, inline-style burn-down, PurgeCSS, perf	P2	S20	21	84	Low	FE2	Not started	Lighthouse, bundle size
S22	Hardening & UAT	Cross-browser, regression, sign-off, contribution prep	P0	all	21	84	Med	all	Not started	Full regression
	Total				464	1,572				

13. Component Effort Estimates

13.1 Per component

Component	Hours	Confidence	Basis
Foundation + self-hosting	104	Medium	Depends on obtaining fonts/sprites outside the CDN
Design tokens	112	High	Token set fully enumerated in §11
Buttons	8	High	Class-identical
Forms (control/select/check/IG)	88	Medium	3,369 controls, ~40 dense forms
Validation	72	Medium	Plugin retained; states re-skinned
Cards	4	High	Class-identical
Tables + DataTables	136	Low	101 pages × 2 modes; Yajra coupling
Navbar / Header	84	Medium	195 !important to unwind
Sidebar	136	Low	316 !important + frozen RBAC contract
Breadcrumb / Tabs / Pagination	64	High	Small, well-bounded
Overlays (7 components)	72	High	All class-identical
Stepper / Search / Chips	72	Medium	Net-new UX4G components
Plugins & dead-code purge	84	Medium	1.4 MB removal is mechanical
Icon consolidation	64	Low	4,011 refs, 2 systems with no confirmed stylesheet
Accessibility	112	Medium	§10.2 itemised (incl. 3 failing UX4G button variants)
Custom widgets	84	Low	Calendar CSS alone is 62 KB
CSS refactor (§9.4)	408	Low	<code>styles.css</code> reduction is the single biggest unknown
Page sweep	400	Medium	537 views × ~0.75 h
Hardening / UAT	84	Medium	
Date Picker	TBD	None	Component existence unverified (R-2)

13.2 Per module — page sweep

Estimated at **0.75 h/page** blended (simple index 0.5 h; form/wizard 1.5 h).

Module	Blades	Hours
<code>mess</code>	76	57
<code>estate</code>	48	36
<code>master</code>	42	32
<code>security</code>	39	29
<code>forms</code>	31	23
<code>course-repository</code>	27	20
<code>registration</code>	23	17
<code>feedback</code>	21	16
<code>setup</code>	18	14
<code>dashboard</code>	17	13
<code>courseAttendanceNoticeMap</code>	16	12
<code>member</code>	15	11
<code>fc-activities</code>	15	11
<code>calendar</code>	15	11
<code>user_management</code>	13	10

Module	Blades	Hours
issue_management	12	9
faculty	12	9
programme	7	5
memo_discipline	7	5
mdo_escrot_exemption	7	5
leave	7	5
25 smaller modules (3–6 each)	~110	83
FC portal	31	23
Total	~608	~456

Note: module Blade counts include partials, so the 608 total exceeds the 537 @extends count. The 400 h figure in §12/§13.1 uses the 537 page-view basis; the 456 h here is the partial-inclusive upper bound. Plan to **440 h ± 15 %**.

13.3 Whole-application rollup

Scenario	Hours	Calendar (3 FE)	Confidence
Optimistic (P10)	1,180	~4.0 months	Low
Expected (P50)	1,572	~5.2 months	Medium
Pessimistic (P90)	2,100	~7.0 months	Medium

The P10→P90 spread is driven almost entirely by two variables: how much of styles.css survives, and whether R-1 forces a full re-theme.

13.4 Team

Role	FTE	Hours	Notes
Frontend developers	3.0	1,572	Sequenced per §12
QA engineer	1.0	~480	≈ 31 % of dev — high because 494 pages need visual UAT
UI/UX designer	0.4	~200	Token decisions, component gallery, UAT sign-off
Accessibility specialist	0.2	~100	S14 + audit; can be the QA engineer if certified
Tech lead / architect	0.3	~150	Reviews, RBAC contract guardianship
Total	4.9 FTE	~2,470 person-hours	~5.2 months

14. Risk Matrix

ID	Risk	Prob.	Impact	Score	Mitigation
RK-1	Only <code>UX4G@2.0.8</code> resolves on the CDN; 2.1.0 / 3.0.0 / <code>latest</code> all 404 while docs advertise v3.0	Confirmed	High	●	Pin 2.0.8 and self-host. Formally request the v3.x distribution from NeGD before S1. Treat v3 (Web Components, <code>ux4g-*</code> prefix) as a separate future project , not this one.
RK-2	UX4G CDN fonts and icon sprites 404 — Noto Sans and all 5 gov sprite sheets are unreachable	Confirmed	High	●	Self-host mandatory (S1). Source Noto Sans from Google Fonts; escalate sprites to NeGD.
RK-3	7,128 <code>!important</code> rules silently defeat UX4G tokens	Confirmed	High	●	<code>@layer</code> architecture (C1) neutralises most without editing them; systematic burn-down in S21.
RK-4	Sidebar work breaks the RBAC nav resolver	High	Critical	●	Contract-freeze the resolver. CSS/markup-only changes in S9. Full role-matrix regression.
RK-5	DataTables theme breaks across 101 pages	High	High	●	S7 is dedicated. Never convert a Yajra table to custom DOM/colVis. Test server-side and client-side separately.
RK-6	Brand-colour decision (R-1) reverses after S2	Medium	Critical	●	Force the decision before S1. Isolating all colour in <code>ux4g-tokens.css</code> makes reversal a one-file change.
RK-7	Icon consolidation regresses 4,011 icons; FA (169) + Iconify (397) have no confirmed stylesheet	High	Medium	●	Triage in S0. Keep all icon fonts loaded until the sweep completes.
RK-8	Three Bootstrap versions cause misattributed bugs	Confirmed	Medium	●	S0 unification is a hard gate.
RK-9	jQuery served only from <code>code.jquery.com</code> (local file is 0 bytes) — CDN outage = total frontend failure	Confirmed	Critical	●	Restore the local copy in S0. This is a live production risk today, independent of UX4G.
RK-10	UX4G Date Picker does not exist in a usable form	Medium	Medium	●	Keep daterangepicker. Do not schedule removal until verified.
RK-11	2,956 inline <code>style=</code> attributes resist tokenisation	Confirmed	Medium	●	Opportunistic burn-down in S16–S20; accept residue.
RK-12	379 inline <code><style></code> blocks + <code>CompressResponse</code> produce blank pages (known failure mode)	Medium	High	●	Escape Blade directives in comments as <code>{{-- @section --}}</code> ; add a smoke test asserting non-empty gzip response per layout.
RK-13	Client-side rendering of large tables blows page weight (precedent: 318 KB → 17 MB on 6,328 rows)	Medium	High	●	Never drop server pagination during restyling. Measure page weight in S7 acceptance.
RK-14	Timetable layout diverges (own Bootstrap, own CSS)	Confirmed	Low	●	Fold into <code>admin.layouts.master</code> in S8, or migrate separately.
RK-15	UX4G <code>--bs-font-monospace: 'Noto Sans'</code> — not a monospace font	Confirmed	Low	●	Override in <code>ux4g-tokens.css</code> .
RK-16	Non-Gol CDN dependencies fail a GIGW audit	Confirmed	Medium	●	Self-hosting in S1 resolves; also removes <code>pinimg.com</code> / <code>codepen.io</code> / <code>wikimedia.org</code> references.
RK-17	494-page visual UAT exceeds QA capacity	Medium	High	●	Automate with Playwright screenshot diffing (baseline in S0); reserve manual QA for the ~120 highest-traffic pages.
RK-18	UX4G 2.0.8 is unmaintained / Gol pivots to v3 mid-project	Medium	High	●	Self-hosting insulates from upstream change. The <code>@layer</code> + token architecture makes a future v3 swap a foundation-layer task, not a rewrite.

15. Rollback Strategy

15.1 Design principle

Because UX4G is class-compatible with Bootstrap, **rollback never requires reverting Blade markup**. Every stage is reversible by changing asset links, not templates. This is the single greatest de-risking property of this migration and should be protected — *do not introduce `ux4g-*`-prefixed classes into Blade*, as that would forfeit it.

15.2 Levels

Level	Trigger	Action	Time to recover	Data loss
L0 — Kill switch	Any production incident	Feature-flag the stylesheet: <code>@if(config('ui.ux4g')) ux4g @else styles.css @endif</code> in <code>pre_header.blade.php</code>	< 2 min (config change)	None
L1 — Token revert	Visual regression from S2	Revert <code>ux4g-tokens.css</code> to previous commit	< 5 min	None
L2 — Component revert	One component regresses	Restore that component's override block in <code>sargam-app.css</code>	< 30 min	None
L3 — Sprint revert	A sprint fails UAT	<code>git revert</code> the sprint's merge commit	< 2 h	None
L4 — Full rollback	Programme cancelled	Delete UX4G <code><link></code> / <code><script></code> ; restore original 14-file load order	< 1 day	None — Blade markup was never changed

15.3 Required controls

- Feature flag from day one.** `config/ui.php` → `'ux4g' => env('UI_UX4G', false)`. Sprint 1's first deliverable.
- Branch per sprint**, merged to a long-lived `feature/ux4g` integration branch. Never migrate directly on `main`.
- Never delete a stylesheet in the same sprint that replaces it.** Comment out the `<link>`, delete the file one sprint later. This makes L2/L3 rollback trivial.
- Tag before each sprint merge:** `ux4g-s{N}-pre`.
- Do not co-mingle** backend changes into migration commits — this preserves `git revert` as a clean operation and honours the no-backend-change constraint.

16. Testing Strategy

16.1 Existing capability

Playwright is already installed (`@playwright/test ^1.58.2` , `playwright.config.js` , `playwright-report/` , `test-results/`). **Needs Code Inspection** — the existing spec coverage was not enumerated in this audit. Extend rather than replace.

16.2 Layers

Layer	Tool	Scope	Gate
Visual regression	Playwright screenshot diff	Baseline 60 pages in S0 → 200 pages by S16	Every PR
Component gallery	New Blade route <code>/ui/gallery</code> (dev-only)	All 30 mapped components, every variant/state	Manual review per sprint
Functional E2E	Playwright	40 critical forms, 207 modals, 101 DataTable pages, all wizards	Per sprint
RBAC matrix	Playwright, one run per role	Sidebar + tab resolution per role — mandatory gate for S9	S9, S22
Accessibility (automated)	<code>axe-core</code> via Playwright	All 537 views	Every PR from S14
Accessibility (manual)	NVDA + keyboard-only	40 representative pages	S14, S22
Contrast	Automated sweep post-S2	All token pairs, 600 badges, 384 alerts	S2, S14
Cross-browser	Playwright (Chromium, Firefox, WebKit) + manual Edge	Top 60 pages	S22
Responsive	Playwright at 576/768/992/1200/1400	Top 60 pages	S22
Performance	Lighthouse CI + bundle analysis	Page weight, CSS size, LCP	S21
Payload smoke	Assert non-empty gzip response per layout	All 5 layouts	Every PR (guards RK-12)
Page weight guard	Assert response < 2 MB on index routes	All list pages	S7 onward (guards RK-13)

16.3 Acceptance criteria per sprint

A sprint is **Done** when: visual diff shows only *intended* changes; zero new axe-core violations; all E2E green; page weight has not regressed; rollback verified by actually toggling the feature flag on staging.

16.4 UAT

- **S16–S20:** module owners UAT their own module against a checklist (layout, forms submit, tables paginate/filter/export, modals, print/PDF/Excel exports unchanged).
- **S22:** full-portal UAT with LBSNAA stakeholders, incl. accessibility sign-off against GIGW 3.0.
- **Export parity is a hard gate** — Print, PDF and Excel exports must keep identical headers and columns. Restyling must not alter export output.

17. Contribution Candidates for UX4G

UX4G is explicitly positioned as an evolving, shared government component ecosystem (MIT-licensed, NeGD/MeitY). Sargam 2.0 contains several components with genuine cross-department reuse value. Ranked by contribution value.

17.1 Ranked candidates

#	Component	Source	Reuse value	Effort to generalise
1	Role-based Dynamic Navigation	<code>SideBarNavResolver</code> , <code>MenuRouteMatcher</code> , <code>components/menu/*</code> (24 files)	Very High	80 h
2	Government Data Table chrome	<code>datatable-global-ui.js</code> (23.6 KB), <code>programme-dt-*</code> pattern	Very High	60 h
3	Approval Workflow Stepper	<code>estate-workflow-stepper.blade.php</code> , <code>partials/step-indicator</code>	High	40 h
4	Weekly Timetable / Schedule Grid	<code>WeeklyTimetableBuilder</code> , 3 export routes	High	64 h
5	Branded Export Toolkit (Print/PDF/Excel)	Export classes + <code>WithEvents/AfterSheet</code>	High	56 h
6	Government Dashboard Widgets	<code>dashboard-stat-cards.css</code> , <code>dashboard-main.css</code>	Medium-High	32 h
7	ID Card / Pass Generator	<code>employee_idcard</code> , <code>family_idcard</code> , <code>duplicate_vehicle_pass</code>	Medium	48 h
8	Institutional Calendar	<code>calendar-admin.css</code> (62 KB), <code>cal-event-pill.css</code>	Medium	72 h
9	Global Success Toast	<code>sargam-success-toast.js</code>	Medium	16 h
10	Faculty / Person Card	<code>components/profile</code> , <code>guest_faculty.css</code> , <code>inhouse_faculty.css</code>	Low-Medium	24 h

17.2 Detail on the top three

C-1 · Role-Based Dynamic Navigation ★ strongest candidate

- **Why reusable.** Every government portal has RBAC-driven navigation. This implementation resolves the active tab, category, group and menu across server and client from a database-backed menu tree with `spatie/laravel-permission` — a problem each department currently re-solves. UX4G ships a Navbar and an Offcanvas but **no navigation-state resolution**, which is where all the real complexity lives.
- **Generalisation needed.** Extract the resolver behind an interface; decouple from Sargam's 5-tab pane structure (§2.2) and from LBSNAA table names; publish as a framework-agnostic JSON menu contract + a reference Blade/React renderer; ship migrations for the menu schema.
- **Dependencies.** Laravel + spatie/permission for the reference implementation; the *contract* is framework-neutral.
- **Accessibility.** Already carries `aria-expanded` (210 sites). To contribute it needs: full `role="navigation"` landmarks, roving-tabindex keyboard traversal, `aria-current="page"`, and screen-reader announcement of tab changes.
- **Effort.** 80 h (40 extract/decouple, 16 a11y hardening, 16 docs/Storybook, 8 tests).

C-2 · Government Data Table Chrome ★ highest immediate impact

- **Why reusable.** Data-heavy tables are the dominant UI of government administration. This codebase has 101 DataTable pages and a mature shared chrome: filter-pill toolbar, column-visibility control, "Showing N of M items" summary, reset-filters, and branded Print/PDF/Excel export. UX4G has **no table component at all** — this is a genuine gap in the design system.
- **Generalisation needed.** Decouple from Yajra (support both server-side and client-side); replace the bespoke filter pills with UX4G **Chips** (which already exist — a natural fit); make the export branding configurable per department; extract `programme-dt-*` into a documented `ux4g-datatable` class contract.
- **Dependencies.** DataTables.net (MIT) — acceptable; would need to be declared as an optional peer dependency, or the chrome offered as adapter-based.
- **Accessibility.** Needs the most work: `aria-live` for async updates (currently absent — gap A-3), `<caption>`, `scope` attributes, sortable-header `aria-sort`, and keyboard-operable column-visibility and filter chips.
- **Effort.** 60 h (24 decouple, 20 a11y, 16 docs).

C-3 · Approval Workflow Stepper

- **Why reusable.** Multi-stage approval is universal in government (leave, exemption, procurement, estate). UX4G ships a **Stepper** (25 CSS classes incl. `.stepper-vertical`, `.stepper-mobile`, `.stepper-invalid`, `.stepper-warning`) but it is presentational — it models *steps*, not *approvals*. Sargam's version carries approver identity, timestamps, remarks and rejection paths.
- **Generalisation needed.** Best contributed as an **extension of the existing UX4G Stepper** rather than a competing component: add `.stepper-approval` states (pending / approved / rejected / returned) plus an approver-metadata slot. This maximises the chance of acceptance upstream.
- **Dependencies.** UX4G Stepper (already present) — no new dependency. Retires `jquery-steps`.

- **Accessibility.** Needs `aria-current="step"`, an ordered-list structure, and a `visually-hidden` status text per step so screen-reader users get state without colour.
- **Effort.** 40 h (16 generalise as an extension, 12 a11y, 12 docs/tests).

17.3 Recommended contribution approach

1. **Contribute nothing until Sprint 22.** Contributing components mid-migration means maintaining a public API while it is still churning.
 2. **Lead with C-2 (Data Table).** It fills a confirmed hole in UX4G's component set, which makes acceptance most likely and the goodwill most useful.
 3. **Contribute C-3 as an extension**, not a replacement — upstream projects accept additive PRs far more readily.
 4. **Budget 216 h** for the top three, scheduled as a post-migration Sprint 23–24. Do not fold it into the migration estimate; it is a separate initiative with its own approval path through NeGD.
-

18. Final Recommendations

18.1 The strategic reframe

Do not scope this as a design-system migration. Scope it as a CSS debt-repayment programme that happens to land on UX4G.

The evidence is unambiguous: UX4G v2.0.8 is Bootstrap 5.3 with a government palette, Noto Sans, and two extra components (Chips, Stepper). Its JS registers the global as `bootstrap`, fires `.bs.*` events, and reads `data-bs-*` attributes. Its CSS uses `--bs-*` custom properties. **Not one of the 853 Blade files needs its classes renamed.**

What actually consumes the 1,572 hours is:

Cost driver	Share
CSS refactor (~1.7 MB, 7,128 <code>!important</code>)	27 %
Page-by-page visual QA (537 views)	26 %
Chrome rebuild (header + sidebar, 511 <code>!important</code>)	14 %
DataTables + forms reconciliation	15 %
Accessibility remediation	7 %
Icons, plugins, widgets, cleanup	11 %

Every one of those is work the project needs regardless of UX4G. **UX4G is the forcing function, not the cost.** Frame it that way to the steering committee — it converts a compliance expense into a technical-debt investment with a compliance dividend.

18.2 Do these seven things

- Resolve R-1, R-2 and R-3 before Sprint 1.** All three are now answered with measured evidence in the companion **Decision Note** (`docs/UX4G-Decision-Note-R1-R2-R3.md`): retain LBSNAA navy `#004384` (9.83:1 AAA vs UX4G violet's 6.12:1 AA) via a compiled override — *not* a token override, because UX4G hard-codes its brand colour; pin `UX4G@2.0.8` (verified as the newest version the CDN actually serves); and self-host, which is mandatory rather than optional because the CDN serves no fonts and no icon sprites at any version.
- Ship Sprint 0 regardless of the UX4G decision.** Three Bootstrap versions, a 0-byte local jQuery served from `code.jquery.com`, and two icon systems with no confirmed stylesheet are live production risks today. Fix them whether or not the migration is approved.
- Adopt `@layer` on day one.** This is the highest-leverage decision available: it neutralises most of the 7,128 `!important` declarations without editing them, converting the largest cost driver from "reason about each rule" to "delete in bulk later".
- Feature-flag from the first commit** and never change Blade classes to `ux4g-*`. Together these keep full rollback at under two minutes for the entire programme.
- Freeze the RBAC sidebar contract.** `SidebarNavResolver` and the 5-pane tab logic in `master.blade.php:62-91` are the most fragile code in the frontend. CSS and markup may change; resolution logic may not. Make this a written rule and a PR-review checklist item.
- Automate visual QA in Sprint 0.** 537 pages cannot be manually verified repeatedly. A Playwright baseline captured *before* any UX4G work is the difference between a 5-month and a 7-month programme.
- Delete 1.4 MB of unused JavaScript.** TinyMCE, Quill, jVectorMap (a US county map), Owl Carousel, Inputmask, Prism, noUiSlider, Typeahead and jQuery UI have **zero** Blade references. This is free performance, and it should be its own small PR in Sprint 0 to build early momentum.

18.3 What to tell the steering committee

The portal can be made UX4G-compliant without changing any backend code, any route, any API or any database object — and without rewriting a single component's markup, because UX4G is class-compatible with the Bootstrap version already in use. The programme is approximately five months with three frontend developers and one QA engineer. Roughly half the effort repays existing CSS debt that is already degrading maintainability and accessibility, so the compliance outcome is largely a by-product of work the portal needs anyway. Two upstream defects — UX4G's CDN does not serve its own fonts or government icon sprites, and only version 2.0.8 resolves despite the documentation advertising 3.0 — require self-hosting and should be raised with NeGD.

18.4 Items requiring confirmation before Sprint 1

Item	Why it is unresolved
UX4G Date & Time Picker	Documented as a component; no <code>.datepicker</code> / <code>.timepicker</code> classes found in <code>ux4g-min.css</code> . May be v3-only or JS-only.
UX4G tonal button variants	Documented; exact class names not published.
UX4G focus-ring and elevation tokens	Not present in the <code>:root</code> block inspected.

Item	Why it is unresolved
UX4G v3.0 availability	CDN returns 404 for all 3.x paths.
Iconify runtime (397 refs)	No Iconify script found in <code>admin_assets/libs/</code> .
Font Awesome stylesheet (169 refs)	Not present in any layout's load order.
jQuery UI (255 KB)	No direct references; may be a transitive dependency.
Existing Playwright spec coverage	Config and reports exist; specs not enumerated in this audit.
Custom 5-pane tab ARIA semantics	<code>role="tablist"</code> / <code>aria-selected</code> usage unverified.
Contrast ratios for 600 badges / 384 alerts	Not measured; requires automated sweep post-token change.
npm package name for UX4G	Not published in the Getting Started documentation.

Document status: Complete. All quantitative claims are measured from the working tree at HEAD `60715da05` or from `cdn.ux4g.gov.in` artefacts downloaded during this audit. Unconfirmed items are enumerated in §18.4 and marked *Needs Code Inspection* inline.

Sources

- [UX4G Design System 3.0](#)
- [UX4G Developer Documentation](#)
- [UX4G Components](#)
- [UX4G Getting Started](#)
- [UX4G Design System Documentation](#)
- [UX4G Design System for Developers and Designers](#)